

Sparse control of alignment models in high dimension

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Overview

- 1 Introduction
- 2 Johnson-Lindenstrauss embeddings
- 3 Analysis on dimension reduction dynamics
- 4 Analysis on control systems
 - Definition of control function
 - Dimension reduction error estimates on Control systems
 - From low-dimension to high-dimension Flocking
- 5 Construction of J-L Matrix for control systems

Goal of this paper

Theorem [Johnson-Lindenstrauss Lemma]

For given $\mathcal{P} \subset \mathbb{R}^d$ with $|\mathcal{P}| < \infty$, there exists $k \leq d$ and $M \in \mathbb{R}^{k \times d}$ such that

$$(1 - \varepsilon)\|x - x'\| \leq \|Mx - Mx'\| \leq (1 + \varepsilon)\|x - x'\|,$$

for all $x, x' \in \mathcal{P}$ where $k \sim \varepsilon^{-2} \log |\mathcal{P}|$.

The main goal of this paper "Sparse control of alignment models in high dimension [Bongini, Fornasier, Junge, Scharf]" is to reduce the space dimension d .

Meta-theorem

For high dimensional particle systems, one can construct **low-dimensional representations** of the dynamical system, which allow the learning of **nearly optimal control strategies** in high dimension with overwhelming confidence.

Cucker-Smale model

For $d \gg 1$ dimensional Cucker-Smale model, $(x, v) \in \mathbb{R}^{2d}$,

$$(\text{Dynamics 1}) \begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = \frac{1}{N} \sum_{j=1}^N a(\|x_i - x_j\|)(v_j - v_i), \quad i = 1, \dots, N, \end{cases}$$

Consider a low-dimensional representation $y_i = Mx_i$, $w_i = Mv_i$,

$$(\text{Dynamics 2}) \begin{cases} \dot{y}_i = w_i \\ \dot{w}_i = \frac{1}{N} \sum_{j=1}^N a(\|y_i - y_j\|)(w_j - w_i), \quad i = 1, \dots, N, \end{cases}$$

with initial data $(y(0), w(0)) = (Mx(0), Mv(0))$.

- ① How small should k be?
- ② How can we construct matrix M with J-L property?
- ③ How close are they?

Cucker-Smale model

For $d \gg 1$ dimensional Cucker-Smale model, $(x, v) \in \mathbb{R}^{2d}$,

$$(\text{Control System 1}) \begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = \frac{1}{N} \sum_{j=1}^N a(\|x_i - x_j\|)(v_j - v_i) + u^h(t). \end{cases}$$

Consider a low-dimensional representation $y_i = Mx_i$, $w_i = Mv_i$,

$$(\text{Control System 2}) \begin{cases} \dot{y}_i = w_i \\ \dot{w}_i = \frac{1}{N} \sum_{j=1}^N a(\|y_i - y_j\|)(w_j - w_i) + u^l(t) \end{cases}$$

with initial data $(y(0), w(0)) = (Mx(0), Mv(0))$.

$u^l(t)$ and $u^h(t)$ will be defined by the information of *(Control System 2)*.

Cucker-Smale model

Goal

- ① $\|y - Mx\| + \|w - Mv\| \leq \varepsilon e^{Ct}$ for $t < T$, both in (*Dynamics*) and (*Control System*).
- ② From the Sparse control, both (*Control System* 1, 2) satisfy flocking condition in finite time, which determined by (*Control System* 2) only.
- ③ $k \sim e^N [T \log N + \log T + \log d]$.

Sections

1. Flocking condition of C-S model,
2. Continuous J-L embedding,
3. Analysis on (*Dynamics*),
4. Analysis on (*Control System*),
5. Constrution of J-L embedding,
6. Numerical experiments.

Johnson-Lindenstrauss property

Definition 2.1

$M \in \mathbb{R}^{k \times d}$, $\delta > 0$, $\varepsilon \in (0, 1)$. M has weak J-L property of ε, δ at $x \in \mathbb{R}^d$

$$\begin{aligned} \text{if either } & (1 - \varepsilon)\|x\| \leq \|Mx\| \leq (1 + \varepsilon)\|x\|, \\ & \text{or } \|x\| \leq \delta \text{ and } \|Mx\| \leq \delta. \end{aligned}$$

M has strong J-L property if it satisfies the first inequality only.

Remark 2.2

The earliest result provides $k \sim \varepsilon^{-2} \log N$ where N is the number of sampling points x .

Johnson-Lindenstrauss property

Remark

Construction of M can be done as a random matrix, then M satisfy strong J-L property with a high probability.

(S1) $M = (m_{ij})$ with $m_{ij} \sim \mathcal{N}(0, 1/k)$.

(S2) $M = (m_{ij})$, and $m_{ij} = \pm 1/\sqrt{k}$ with probability $1/2$. Then

$$1 \leq \|M\|_{l_2 \rightarrow l_2} \leq \sqrt{d}$$

(S3) $M = (m_{ij})$ is random projections (choose random k orthogonal vectors and project to it) scaled by factor $\sqrt{d/k}$. Then

$$1 \leq \|M\|_{l_2 \rightarrow l_2} \leq \sqrt{d/k}$$

In particular, (S3) is stated in [12], and we will use this.

Johnson-Lindenstrauss property

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In particular, (S3) is stated in [12], and we will use (S2) or (S3).

Johnson-Lindenstrauss property

Remark 2.3 : Continuous J-L property

Given $\varepsilon > 0$ and a C^1 -curve $\phi : [0, 1] \rightarrow \mathbb{R}^d$, if $\rho := \max_{t \in [0, 1]} \frac{\|\phi'(t)\|}{\|\phi(t)\|} < \infty$, then there exists $M \in \mathbb{R}^{k \times d}$ for $k \sim \varepsilon^{-2} \log(d \cdot \rho \cdot \varepsilon^{-1})$ such that

$$(1 - \varepsilon)\|\phi(t)\| \leq \|M\phi(t)\| \leq (1 + \varepsilon)\|\phi(t)\| \text{ for all } t \in [0, 1].$$

Note

We want $\phi(t) := x_i - x_j$ or $v_i - v_j$, but this result need condition $\rho < \infty$, which is impossible if collision occurs.

This is why we need weak J-L property.

Johnson-Lindenstrauss property

Lemma 2.4 : Continuous J-L property

Let $\phi : [0, 1] \rightarrow \mathbb{R}^d$ Lipschitz function with constant $L_\phi = L_\phi(0, 1)$ and let $\delta > 0, \varepsilon \in (0, 1)$. Suppose $M \in \mathbb{R}^{k \times d}$ has strong J-L property of $\tilde{\varepsilon} = \varepsilon/2$ at $\mathcal{N}$ arbitrary points with a high probability where $\mathcal{N} \geq 4 \cdot \frac{L_\phi(\sqrt{d}+2)}{\delta\varepsilon}$. Then M has weak J-L property of ε, δ at $\phi(t)$ with the same probability.

Proof

Use Remark 2.3. Let $t_i := i/\mathcal{N}$ for $i = 0, \dots, \mathcal{N} - 1$.

From assumption, M satisfies J-L at $\phi(i/\mathcal{N})$ with a high probability, and $1 \leq \|M\| \leq \sqrt{d}$.

Let $t \in [0, 1]$ and choose j s.t. $t \in [t_j, t_{j+1}]$.

Then there are two cases, one is

$$\|\phi(t_j)\| \leq \delta/2,$$

and the other is

$$\|\phi(t_j)\| > \delta/2.$$

Johnson-Lindenstrauss property

Proof (Continued)

(Case 1) $\|\phi(t_j)\| \leq \delta/2$.

Since $\varepsilon \in (0, 1)$ and $\mathcal{N} \geq 4 \cdot \frac{L_\phi(\sqrt{d}+2)}{\delta\varepsilon}$,

$$\|\phi(t)\| \leq \|\phi(t) - \phi(t_j)\| + \|\phi(t_j)\| \leq L_\phi/\mathcal{N} + \delta/2 \leq \delta, \quad (\text{Lipschitz})$$

$$\begin{aligned} \|M\phi(t)\| &\leq \|M\|\|\phi(t) - \phi(t_j)\| + \|M\phi(t_j)\| \leq \sqrt{d}L_\phi/\mathcal{N} + (1 + \varepsilon')\|\phi(t_j)\| \\ &\leq \delta/4 + 3/2 \cdot \delta/2 \leq \delta. \end{aligned}$$

(Case 2) $\|\phi(t_j)\| > \delta/2$.

$$\begin{aligned} \|\phi(t) - \phi(t_j)\| &\leq \frac{L_\phi}{\mathcal{N}} \cdot \frac{2\|\phi(t_j)\|}{\delta} \leq \frac{\delta\varepsilon}{4(\sqrt{d}+2)} \cdot \frac{2\|\phi(t_j)\|}{\delta} \\ &\leq \|\phi(t_j)\| \cdot \frac{(\varepsilon - \tilde{\varepsilon})}{\|M\| + 1 + \varepsilon} =: \tau. \end{aligned}$$

Johnson-Lindenstrauss property

Proof (Continued)

$$\begin{aligned}
 \|\phi(t) - \phi(t_j)\| &\leq \|\phi(t_j)\| \cdot \frac{(\varepsilon - \tilde{\varepsilon})}{\|M\| + 1 + \varepsilon} =: \tau, \\
 \frac{\|M\phi(t)\|}{\|\phi(t)\|} &= \frac{\|M(\phi(t) - \phi(t_j)) + M\phi(t_j)\|}{\|\phi(t) - \phi(t_j) + \phi(t_j)\|} \\
 &\leq \frac{\|M\|\|\phi(t) - \phi(t_j)\| + (1 + \tilde{\varepsilon})\|\phi(t_j)\|}{\|\phi(t_j)\| - \|\phi(t) - \phi(t_j)\|} \\
 &\leq \frac{\|M\|\tau + (1 + \tilde{\varepsilon})\|\phi(t_j)\|}{\|\phi(t_j)\| - \tau} \leq (1 + \varepsilon) \cdot \frac{\|\phi(t_j)\|}{\|\phi(t_j)\| - \tau} \leq 1 + \varepsilon.
 \end{aligned}$$

Now we have matrix $M \in \mathbb{R}^{k \times d}$ which has continuous J-L property with a high probability, and its dimension is

$$k \sim \left(\frac{\varepsilon}{2}\right)^{-2} \log \mathcal{N} \sim \left(\frac{\varepsilon}{2}\right)^{-2} \log \left[4 \frac{L_\phi(\sqrt{d} + 2)}{\delta \varepsilon} \right].$$

Johnson-Lindenstrauss property

Lemma 2.5 : Estimates on variation

Suppose $a_1, \dots, a_N \in \mathbb{R}^d, b_1, \dots, b_N \in \mathbb{R}^k, M \in \mathbb{R}^{k \times d}$ such that $\Delta > 0$ satisfies

- (i) M : weak J-L property with $\varepsilon = 0.5, \delta = \Delta$ for all a_i ,
- (ii) $\|Ma_i - b_i\| \leq \Delta$ for all i .

Let $\hat{i}$ be the smallest index s.t. $\|b_{\hat{i}}\| \geq \|b_j\|$ for all j and let $A := \frac{1}{N} \sum_{j=1}^N \|a_j\|^2, B := \frac{1}{N} \sum_{j=1}^N \|b_j\|^2$.

If $\sqrt{B} \geq 2\Delta$, then for $c = \frac{1}{\sqrt{289}}$,

$$\|a_{\hat{i}}\| \geq \|b_{\hat{i}}\|/4, \|a_{\hat{i}}\| \geq c\sqrt{A}, \text{ and } B \leq 16NA.$$

If $\sqrt{B} \leq 2\Delta$, then for $C = \sqrt{72}$,

$$\sqrt{A} \leq C\Delta.$$

Dimension reduction

$$(Dynamics\ 1) \begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = \frac{1}{N} \sum_{j=1}^N a(\|x_i - x_j\|)(v_j - v_i), \quad i = 1, \dots, N, \end{cases}$$

$$(Dynamics\ 2) \begin{cases} \dot{y}_i = w_i \\ \dot{w}_i = \frac{1}{N} \sum_{j=1}^N a(\|y_i - y_j\|)(w_j - w_i), \quad i = 1, \dots, N, \end{cases}$$

Define $B(u, \tilde{u}) := \frac{1}{2N^2} \sum_{i,j=1}^N \langle u_i - u_j, \tilde{u}_i - \tilde{u}_j \rangle$, and let

$$X(t) = B(x, x), Y(t) = B(y, y), V(t) = B(v, v), W(t) = B(w, w).$$

Define $u_i^\perp = u_i - \bar{u} = u_i - \frac{1}{N} \sum_{j=1}^N u_j$, then $B(u, u) = \frac{1}{N} \sum_{i=1}^N \|u^\perp\|^2$.

Dimension reduction

Remark 3.1

V, W are decreasing. Therefore,

$$\begin{aligned}\|v_i - v_j\| &\leq \sqrt{2NV(0)}, & \|w_i - w_j\| &\leq \sqrt{2NW(0)}, \\ \|x_i(0) - x_j(0)\| &\leq \sqrt{2NX(0)}, & \|y_i(0) - y_j(0)\| &\leq \sqrt{2NY(0)}.\end{aligned}$$

Definition

Define error functions as follows.

$$\begin{aligned}e_i^x(t) &= \|y_i - Mx_i\|, & e_i^y(t) &= \|w_i - My_i\|, \\ \mathcal{E}^x(t) &= \max_{1 \leq i \leq N} e_i^x(t), & \mathcal{E}^y(t) &= \max_{1 \leq i \leq N} e_i^y(t),\end{aligned}$$

$$\mathcal{E}_2^x(t) = \left(\frac{1}{N} \sum_{j=1}^N (e_j^x(t))^2 \right)^{\frac{1}{2}}, \quad \mathcal{E}_2^y(t) = \left(\frac{1}{N} \sum_{j=1}^N (e_j^y(t))^2 \right)^{\frac{1}{2}}.$$

Dimension reduction

Theorem 3.2 : Analysis on (Dynamics 1, 2)

Suppose M has weak J-L property with ε, δ on $x_i(t) - x_j(t)$ for all $t \in [0, T]$ and i, j .

Let L_a be a Lipchitz constant of function a , set

$$K_1 := L_a \sqrt{NW(0)} \sqrt{2X(0)}, \quad K_2 := 2L_a \sqrt{NW(0)},$$

$$K_3 := \frac{1}{2} L_a \sqrt{NW(0)} \sqrt{2V(0)}.$$

Then

$$\mathcal{E}_2^x(t) + \mathcal{E}_2^v(t) \leq ((\varepsilon K_1 + \delta K_2)t + \varepsilon K_3 t^2) \exp(t \|K\|_{l_1 \rightarrow l_1}),$$

$$\mathcal{E}^v(t) \leq \sqrt{N} \min\{\mathcal{E}_2^x(t) + \mathcal{E}_2^v(t), \|M\| \sqrt{V(t)} + \sqrt{W(t)}\},$$

$$\text{where } K = \begin{bmatrix} 2a(0) & 2L_a \sqrt{NW(0)} \\ 1 & 0 \end{bmatrix}.$$

Dimension reduction

Remark 3.4

The assumption of M can be achieved by Lemma 2.4.

It is sufficient to have M such that

M has J-L property for $x_i(t_m) - x_j(t_m)$, where
 $i, j \in \{1, \dots, N\}$, $m = 0, \dots, [T \cdot \mathcal{N}']$, $t_m = \frac{m}{[T \cdot \mathcal{N}'] - 1}$,

$$\mathcal{N}' \geq 4 \cdot \frac{\sqrt{2NV(0)}(\sqrt{d}+2)}{\delta\varepsilon}.$$

proof

$$L_{\phi_{ij}} = \sup \|(x_i - x_j)'(t)\| = \sup \|v_i - v_j\| \leq \sqrt{2NV(0)}.$$

Therefore we can use Lemma 2.4.

Since $[\mathcal{N}' \cdot T]$ points are needed for each (i, j) pair, we have number of sampling points $\mathcal{N} \sim \sqrt{NV(0)} \cdot \frac{\sqrt{d}}{\delta\varepsilon} \cdot T \cdot N^2$. Therefore we have

$$k \sim \varepsilon^{-2} [\log N + \log T + \log d + \log V(0) + |\log \delta\varepsilon|].$$

Note that $\varepsilon \sim e^N$ implies the resulting k in the beginning of presentation.

Dimension reduction

Remark 3.5

We can erase dependence on d of k , if we allow a little weak estimates on Theorem 3.2.

In that case, we have $k \sim \varepsilon^{-2}[\log N + \log T + \log V(0) + |\log \delta|]$, and get $\mathcal{E}_2^x(t) + \mathcal{E}_2^v(t) \leq ((\varepsilon K_1 + \delta K_2)t + \varepsilon K_3 t^2) \exp(2t\|K\|_{l_1 \rightarrow l_1})$

Remark 3.6

We have $\mathcal{E}^v(t) \leq \sqrt{N} \min\{\mathcal{E}_2^x(t) + \mathcal{E}_2^v(t), \|M\|\sqrt{V(t)} + \sqrt{W(t)}\}$.

On the other hand, if we know the initial condition satisfies flocking condition, then $\mathcal{E}^v(t) \leq \sqrt{N} \left(\|M\|\sqrt{V(t)} + \sqrt{W(t)} \right) \rightarrow 0$ as $t \rightarrow \infty$.

Definition of control function

Consider a sparse control system on (Dynamics 1, 2).

$$(\text{Control System 1}) \begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = \frac{1}{N} \sum_{j=1}^N a(\|x_i - x_j\|)(v_j - v_i) + u^h(t). \end{cases}$$

$$(\text{Control System 2}) \begin{cases} \dot{y}_i = w_i \\ \dot{w}_i = \frac{1}{N} \sum_{j=1}^N a(\|y_i - y_j\|)(w_j - w_i) + u^l(t) \end{cases}$$

with initial data $(y(0), w(0)) = (Mx(0), Mv(0))$.

$u^l(t)$ and $u^h(t)$ will be defined by the information of (Control System 2).

How can we define $u^l(t)$ as a feedback control?

Definition of control function

Define $T_0^c := \inf\{t \in [0, T] : W(t) \leq \Gamma\}$ with $\inf \phi = T$, and let $\hat{i}(t) \in \{1, \dots, N\}$ be the smallest index such that

$$\|w_{\hat{i}}^\perp(t)\| = \max_{1 \leq i \leq N} \|w_i^\perp(t)\|.$$

Then if image of control function is $U = [-\theta, \theta]$, define

$$u_i^l(t) := -\theta(\delta_{i\hat{i}}) \cdot \frac{w_{\hat{i}}^\perp(t)}{\|w_{\hat{i}}^\perp(t)\|},$$

$$u_i^h(t) := -\theta(\delta_{i\hat{i}}) \cdot \frac{v_{\hat{i}}^\perp(t)}{\|v_{\hat{i}}^\perp(t)\|}, \text{ for } t \leq T_0^c,$$

and after T_0^c , define $u_i^l(t) = u_i^h(t) = 0$.

Note that the high-dimension control function $u_i^h(t)$ only depends on (Control System 2), the low-dimension control system.

Definition of control function

Define $T_0^c := \inf\{t \in [0, T] : W(t) \leq \Gamma\}$ with $\inf \phi = T$, and let $\hat{i}(t) \in \{1, \dots, N\}$ be the smallest index such that

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and after T_0^c , define $u_i^l(t) = u_i^h(t) = 0$.

Note that the high-dimension control function $u_i^h(t)$ only depends on (Control System 2), the low-dimension control system.

As you can see... It is not well-defined.

Another problem on control systems

From Remark 3.4 and Lemma 2.4, we could obtain a random matrix M , which satisfies strong J-L condition at $\mathcal{N}$ points with high probability.

It was sufficient to use in Theorem 3.2, since the trajectory was deterministic; M only needs to hold J-L on 'certain' $\mathcal{N}'$ points of trajectory.

In Section 4, it does not work.

Let's fix M . (Control System 2) is generated by M . Since we will choose control function $u^l(t)$ as a feedback control, it depends on M .

Now the dynamics of (Control System 2) depends on M . Therefore, if we use the same technique as before, it implies

M should hold J-L property on possible points depending on M itself.

Definition of control function

Answer ;

Choose sampling time $\tau > 0$ and change u only at time $t = n\tau$ for $n \in \mathbb{N}_0$.

Definition 4.3

Given $\Gamma, \tau > 0$, define $u_i^l(t)$ and $u_i^h(t)$ as a piecewise constant function. Let $T_0^s := \inf_{n \in \mathbb{N}_0} \{n\tau : W(n\tau \leq \Gamma)\}$. For $t \in [n\tau, (n+1)\tau)$, let $\hat{i}(t) \in \{1, \dots, N\}$ be the smallest index such that

$$\|w_{\hat{i}}^\perp(n\tau)\| = \max_{1 \leq i \leq N} \|w_i^\perp(n\tau)\|.$$

Now we define the control functions

$$u_i^l(t) := -\theta(\delta_{i\hat{i}}) \cdot \frac{w_{\hat{i}}^\perp(t)}{\|w_{\hat{i}}^\perp(t)\|},$$

$$u_i^h(t) := -\theta(\delta_{i\hat{i}}) \cdot \frac{v_{\hat{i}}^\perp(t)}{\|v_{\hat{i}}^\perp(t)\|}, \text{ for } t \leq T_0^s.$$

Definition of control function

Remark

- 1 Since the control functions are piecewise constant, we have x, v, y, w as a piecewise analytic functions.
- 2 Since the choice of control functions are restricted by $N \cdot T/\tau$, M needs to satisfy J-L condition on $N \cdot T/\tau \cdot \mathcal{N}$, finite sampling points.
- 3 This sparse control clearly leads $W(t)$ to zero. In this way, we can make $\sqrt{W(t)} \leq \left(\int_{X(0)}^{\infty} a(\sqrt{2Nr}) dr \right) =: \gamma(X(0))$, in the flocking region.
- 4 For optimality, see reference [7].

Dimension reduction error

Proposition 4.4 : Theorem 3.2 on control problem

Let $T > 0, \Delta > 0, k \in \mathbb{N}_0$ with $k \leq d$. Let $\tau > 0$ be a sampling time, $\hat{T} > 0$ such that $\hat{T} + \tau \leq T$.

Suppose $M \in \mathbb{R}^{k \times d}$ satisfy the following Johnson-Lindenstrauss properties,

(JL1) Weak J-L property of $\varepsilon = \varepsilon'$ and $\delta = \min \left(\varepsilon' \frac{\sqrt{2X(0)} + \alpha}{2}, \frac{1}{2} \right)$ at all

points $x_i(t) - x_j(t)$ for $t \in [0, \hat{T} + \tau]$, all i, j ,

(JL2) Weak J-L property of $\varepsilon = \varepsilon'$ and $\delta = \Delta$ at $v_i^\perp(n\tau)$ for all i, n ,

(JL3) Strong J-L property of $\varepsilon = 0.5$ at $v_i^\perp(0)$ and $x_i^\perp(0)$ for all i ,

where we define threshold $\Gamma = (2\Delta)^2$, $\int_0^t \sqrt{2V(s)} ds \leq \alpha$ for all

$t \in [0, \hat{T} + \tau]$, and ε' is sufficiently small that

$$\varepsilon' \sqrt{N} \left(4L_a \sqrt{NV(0)} (\sqrt{2X(0)} + \alpha) + \frac{\theta}{\sqrt{N}} \right).$$

$$(\hat{T} + \tau) \exp \left[(\hat{T} + \tau) \cdot (\max\{2a(0) + 1, 4L_a \sqrt{NV(0)}\} + \frac{8\theta}{\Delta}) \right] \leq \frac{\Delta}{2}.$$

Dimension reduction error

Proposition 4.4 (Continued)

Then for $t \in [0, \min\{\hat{T} + \tau, T_0\}]$, we have

$$\begin{aligned} \mathcal{E}^v(t) + \mathcal{E}^x(t) &\leq \sqrt{N}(\mathcal{E}_2^v(t) + \mathcal{E}_2^x(t)) \\ &\leq \varepsilon' \sqrt{N} \left(4L_a \sqrt{NV(0)} (\sqrt{2X(0)} + \alpha) + \frac{\theta}{\sqrt{N}} \right) \cdot \\ &\quad t \exp \left[t \cdot (\max\{2a(0) + 1, 4L_a \sqrt{NV(0)}\} + \frac{8\theta}{\Delta}) \right] \leq \frac{\Delta}{2}. \end{aligned}$$

From low-dimension to high-dimension Flocking

Theorem 4.5 : Flocking of (Control System 1) by (Control System 2)

Let $\bar{X} := 2X(0) + \frac{2N^2}{c^2\theta^2} V(0)^2$, $\bar{Y} := 2Y(0) + \frac{2N^2}{c^2\theta^2} W(0)^2$,

$\Delta := \min\left(\frac{\gamma(\bar{X})}{C}, \frac{\gamma(4\bar{X})}{2}\right)$, $\hat{T} = \frac{2N}{\theta}(2\sqrt{V(0)} - 2\Delta)$, where C, c are constant defined in Lemma 2.5. ($C = \sqrt{72}, c = 1/\sqrt{289}$).

Let $\tau_0 > 0$ so small that

$$\tau_0(a(0)\sqrt{NV(0)} + \theta) + \tau_0^2 a(0)\theta \leq \frac{\Delta}{4},$$

and let the sampling time $0 < \tau < \tau_0$.

Then under the same condition as Proposition 4.4,

$$\sqrt{V(T_0)} < \gamma(X(T_0)), \text{ where } T_0 = \min\{n\tau \mid \sqrt{W(n\tau)} \leq 2\Delta, n\tau \leq \hat{T} + \tau\}.$$

From low-dimension to high-dimension Flocking

Sketch of proof on Theorem 4.5

There are 3 steps to confirm flocking.

(Step 1) Induction on $n\tau$.

In the range of $t \leq (n+1)\tau$, Prove

$$\textcircled{1} \quad V'(t) \leq -\frac{c\theta}{N} \sqrt{V(t)}, \quad W'(t) \leq -\frac{\theta}{N} \sqrt{V(t)},$$

$$\textcircled{2} \quad X(t) \leq \bar{X}, \quad Y(t) \leq \bar{Y},$$

$$\textcircled{3} \quad \int_0^{n\tau} \sqrt{2V(s)} ds \leq \alpha,$$

if these are true for $t \leq n\tau$.

(Step 2) We use Proposition 4.4 to show $\sqrt{W(\hat{T} + \tau)} < 2\Delta$, so that such T_0 exists.

From low-dimension to high-dimension Flocking

Sketch of proof on Theorem 4.5 (Continued)

(Step 3) $\sqrt{V(T_0)} \leq \gamma(X(T_0))$ and $\sqrt{W(T_0)} \leq \gamma(Y(T_0))$ from assumptions and error estimates. In particular,

$$\sqrt{W(T_0)} \leq 2\Delta \leq \gamma(4\bar{X}) \leq \gamma(\bar{Y}) \leq \gamma(Y_0).$$

Since $\mathcal{E}^v \leq \Delta$ and Lemma 2.5,

$$\sqrt{V(T_0)} \leq C\Delta \leq \gamma(\bar{X}) \leq \gamma(X(T_0)).$$

Remark

From Theorem 4.5, we can control the high-dimension control flocking problem from the low-dimension control flocking problem.

Calculation on low-dimension control system

From Theorem 4.5, we need M satisfying (JL1), (JL2), (JL3).

The construction of M is from Remark 2.2, Lemma 2.4 if there is no control function.

In order to add control function, we calculated the number of possible control path, $P = N^{\hat{T}/\tau+1}$.

Note that in Lemma 2.4, $k \varepsilon^{-2} \log \mathcal{N}$ where $\mathcal{N}$ is the number of sampling points.

In Theorem 4.5, $\varepsilon = \min\{\varepsilon', 0.5\}$ with $\varepsilon' \sim e^{-\hat{T}} \sim e^{-N}$.

To estimate $\mathcal{N}$, we should consider all possible path and all sampling points that we need.

Calculation on low-dimension control system

(JL1) weak J-L along continuous path.

$$\mathcal{N}_1 = P \cdot (\hat{T} + \tau) \cdot \frac{N(N-1)}{2} \cdot 4 \cdot \frac{L_x(\sqrt{d}+2)}{\delta\varepsilon}, \text{ where } L_x \leq \sqrt{2NV(0)}.$$

(JL2) weak J-L along each $n\tau$.

N^{n+1} sampling points for each time $t = n\tau$.

Therefore, $\mathcal{N}_2 = N^{[\hat{T}/\tau]+4}$.

(JL3) strong J-L on the initial data. $\mathcal{N}_3 = 2N$.

Therefore,

$$\mathcal{N} \leq \mathcal{N}_1 + \mathcal{N}_2 + \mathcal{N}_3$$

$$\leq N^{[\hat{T}/\tau]+1}(\hat{T} + \tau) \frac{N(N-1)}{2} 4 \cdot \frac{L_x(\sqrt{d}+2)}{\delta\varepsilon} + N^{[\hat{T}/\tau]+4} + 2N,$$

$$k \sim \varepsilon^{-2} \log \mathcal{N} \sim e^{2N} [(\hat{T}/\tau) \log N + \log(\hat{T}/\tau) + \log d + \log V(0) + |\log \delta\varepsilon|]$$

Calculation on low-dimension control system

Remark 5.1

We can eliminate dependence on d using the same technique of Remark 3.5, weakening the estimates on $\mathcal{E}^\vee$.

Remark 5.3

If we prepare considerable independent family of M_n for each $t = n\tau$, then we don't need (JL1) for all possible path. We can reduce $\mathcal{N}_1$ and the condition on M .

The End